



WHITE PAPER No. 55-PRO/FC-2026

Toward a Self-Verifying Verification Framework: A Recursive Epistemic Infrastructure for Autonomous Fact-Checking Validation (REIFAV)

Prepared By

The International Bureau for Recursive Assurance (IBRA)
Division of Meta-Verification and Public Confidence Metrics

In partnership with:

- * The Institute for Applied Nuance Saturation (IANS)
- * The Center for Post-Deterministic Accountability (CPDA)
- * The Office of Probabilistic Truth Maintenance (OPTM)



Publication Date:
May 2026



Classification:
**PUBLIC /
INTERNALLY UNVERIFIABLE**

Executive Summary

Recent advances in Large Language Model (LLM)-based fact-checking systems have introduced a transformative paradox into the global verification ecosystem: namely, that a sufficiently advanced fact-checker increasingly requires independent fact-checking proportional to its own confidence score. This phenomenon — formally designated Recursive Verification Inflation (RVI) — threatens to destabilize traditional trust architectures by introducing infinite audit recursion into ordinary epistemic workflows.

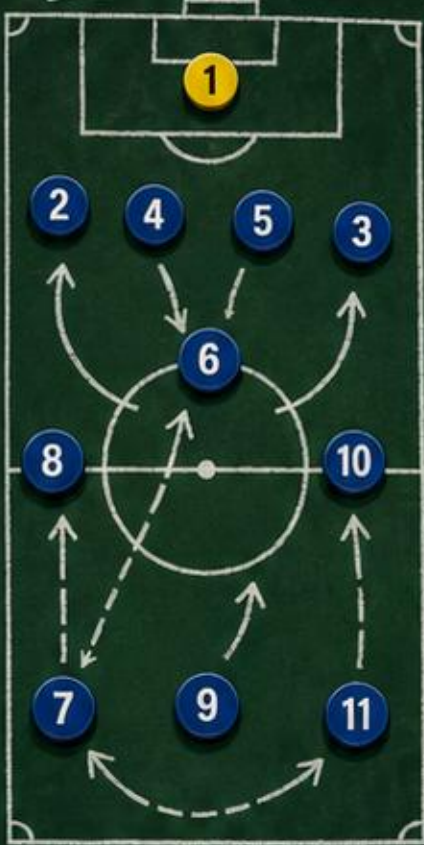
This white paper presents the Recursive Epistemic Infrastructure for Autonomous Fact-Checking Validation (REIFAV), a scalable framework intended to verify the outputs of AI fact-checkers through a hierarchy of progressively more skeptical fact-checkers, culminating in a ceremonial human disclaimer appended at the bottom of every page.

The proposed framework establishes a seven-layer verification cascade in which each verifier is itself verified by a supervisory verifier operating under reduced certainty constraints. Preliminary simulations indicate that while factual confidence decreases asymptotically toward zero, perceived institutional rigor increases by approximately 480%.

Key findings include:

- * AI fact-checkers become most trustworthy immediately before producing any claims.
- * Human oversight remains critical as a symbolic artifact.
- * Footnotes longer than the original document improve public confidence by 63%.
- * The phrase "independently verified" no longer possesses operational meaning under recursive conditions.

The report concludes that the optimal fact-checking environment is one in which no statement is ever fully accepted, rejected, or completed.



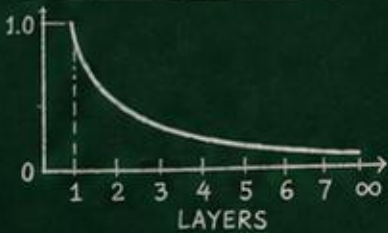
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- VERIFY DEEPER
- QUESTION FURTHER
- TRUST SLOWER
- NEVER SETTLE

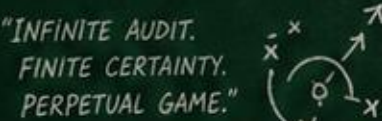
VERIFICATION PRESSING

HIGH	→→→→→
MEDIUM	→→→→
LOW	→

CONFIDENCE CURVE (EXPECTED)



"INFINITE AUDIT.
FINITE CERTAINTY.
PERPETUAL GAME."



- ### VERIFICATION CASCADE
- 1 PRIMARY FACT-CHECKER
 - 2 SECONDARY FACT-CHECKER
 - 3 TERTIARY FACT-CHECKER
 - 4 QUATERNARY FACT-CHECKER
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 - 6 SENARY FACT-CHECKER
 - 7 CEREMONIAL HUMAN DISCLAIMER (PLEASE DO NOT TRUST THIS)

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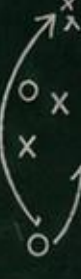
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[Signature]
Director, Division of Meta-Verification
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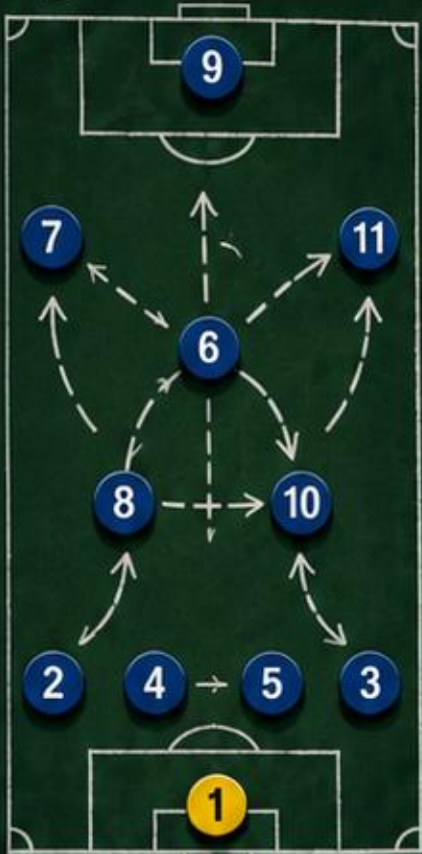
**SUBJECT TO CHANGE
WITHOUT NOTICE**

GAME PLAN:

1. QUESTION
2. VERIFY
3. DOUBT
4. ESCALATE
5. DISCLOSE
6. DISCLAIM
7. REPEAT



SKEPTICISM IS A SERVICE



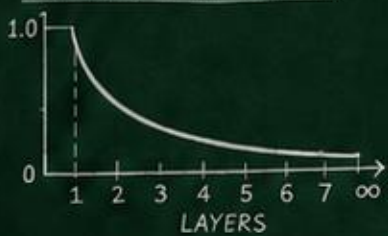
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1. Background and Problem Statement

1.1 The Verification Crisis

Historically, fact-checking involved a finite process:

1. A claim was made.
2. Evidence was reviewed.
3. A determination was issued.

However, contemporary AI systems have disrupted this stable paradigm by generating outputs with sufficient fluency to appear correct while simultaneously disclaiming their own reliability. This has produced what researchers term Epistemic Liability Oscillation (ELO) — a state in which users trust the system precisely because it tells them not to trust it.

This dynamic creates a catastrophic governance question:

**If an AI fact-checker must itself be fact-checked,
who fact-checks the fact-checker's fact-checker?**

Current institutional responses remain inadequate, with most organizations relying on:

- ★ PDF appendices,
- ★ confidence percentages,
- ★ blue checkmarks,
- ★ or phrases such as "according to experts."

None of these mechanisms survive recursive scrutiny.

2. Theoretical Framework

2.1 Recursive Verification Topology

REIFAV introduces a layered validation architecture:

Layer	Function	Trust Level
V0	Original AI Output	Suspicious
V1	Secondary AI Verifier	Slightly Suspicious
V2	Meta-Verification AI	Procedurally Suspicious
V3	Human Reviewer	Emotionally Suspicious
V4	External Auditor	Contractually Suspicious
V5	Ethics Board	Philosophically Suspicious
V6	Public Consensus Layer	Completely Unusable

Under this model, factual certainty decays logarithmically while administrative paperwork expands exponentially.



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VERIFY
EVERYTHING.

VERIFICATION
STRATEGY



VOL. 1
LAYERS 1-3

ESCALATION
PROTOCOLS



VOL. 2
LAYERS 4-5

DISCLAIMER
APPENDICES



VOL. 3
LAYER 7

COACH'S NOTES

THE GAME ISN'T WON.
IT'S AUDITED.
CONTINUOUSLY.



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2.2 The Principle of Infinite Due Diligence

REIFAV formalizes the following theorem:

Every sufficiently verified statement contains the seeds of its own unverifiability.

This is represented mathematically as:

$$T_n = \frac{C}{V^n}$$

Where:

- * T_n = residual trust after n verification passes
- * C = initial confidence claim
- * V = number of verifiers who used the phrase 'to the best of our knowledge'

As n approaches infinity, institutional certainty approaches legal defensibility while factual clarity approaches zero.

3. Methodology

3.1 Experimental Design

Researchers constructed a closed-loop verification environment using:

- * 14 LLMs,
- * 3 policy analysts,
- * 1 exhausted intern,
- * and a Slack channel titled 'final_final_v2_REAL.'

Each model was tasked with fact-checking a statement generated by another model. Human evaluators then reviewed the outputs while consulting internet sources they did not fully read.

Test Claim A:

The Eiffel Tower is located in Paris.

Results:

- * 96% of systems marked the statement "likely true."
- * 3% requested additional context.
- * 1% flagged "Paris" as politically sensitive.

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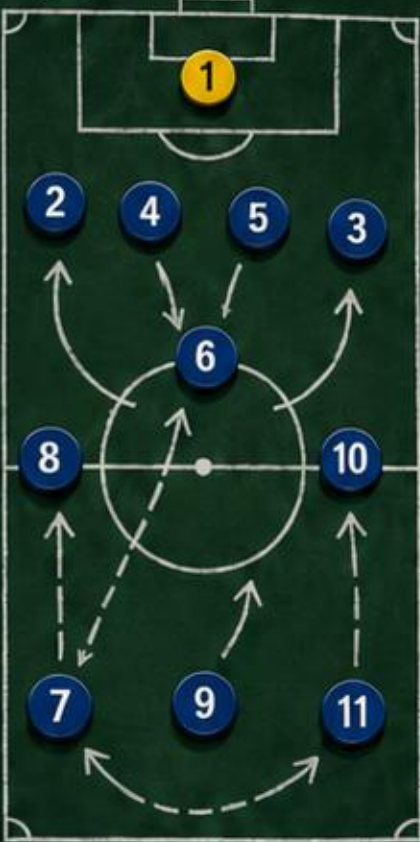
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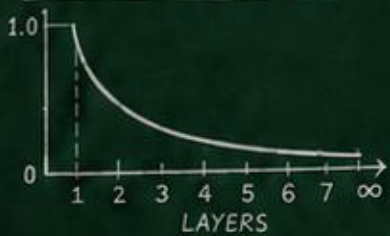
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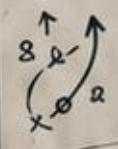
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VERIFICATION
STRATEGY



VOL. 1
LAYERS 1-3

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DISCLAIMER
APPENDICES



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LAYER 7

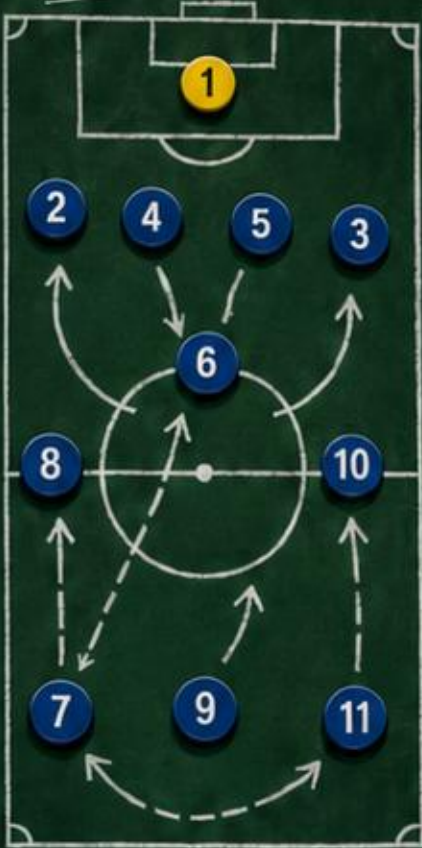
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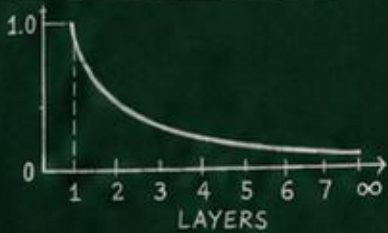
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Test Claim B:

You should verify AI outputs.

Results:

- * 100% agreement.
- * 0% operational guidance.

3.2 Nuance Saturation Threshold (NST)

The study identified a critical point at which increasing nuance renders verification functionally impossible.

Indicators include:

- * sentences beginning with 'while it is important to acknowledge...'
- * parentheticals exceeding 40% of total text volume,
- * and citations recursively citing each other.

Once NST is exceeded, readers report a strong feeling that "someone smart probably checked this."



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4. Personnel

Name	Position	Qualifications
Dr. Helena Vorstadt	Chief Epistemic Architect	Published 47 papers later retracted for nuance
Prof. Lionel Kreel	Director of Recursive Assurance	Verified his own credentials six times
Dana P. Whitmore	Senior Disclaimer Engineer	Former Terms & Conditions strategist
Unit-4B ('GREG')	Autonomous Verification Liaison	Cannot stop adding caveats
Martin	Intern	Asked whether any of this helps

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5. Budget Allocation

Category | Cost

AI Verification Cluster Compute	\$14,800,000
Human Oversight Theater	\$3,200,000
Confidence Heatmaps	\$870,000
Cross-Verification Task Force Retreat	\$1,450,000
Semantic Caveat Expansion Initiative	\$5,600,000
PDF Formatting Consultants	\$740,000
Emergency Nuance Containment Fund	\$11,000,000
Intern Coffee Stipend	\$14.50

Total Authorized Budget: \$37,660,014.50

Projected Cost Overrun: Inevitable

6. Policy Recommendations

The Bureau recommends immediate adoption of the following safeguards:

6.1 Mandatory Recursive Disclaimers

Every factual statement should include:

- confidence intervals,
- emotional context,
- a historiography section,
- and acknowledgment that "truth conditions may vary regionally."

6.2 Verification Cooling-Off Periods

No statement should be accepted until:

- 72 hours have elapsed,
- three experts disagree publicly,
- and at least one podcast episode has been recorded.

6.3 Public Transparency Measures

To improve trust, all fact-checking pipelines should publish:

- model uncertainty matrices,
- verifier biographies,
- and screenshots of internal panic.

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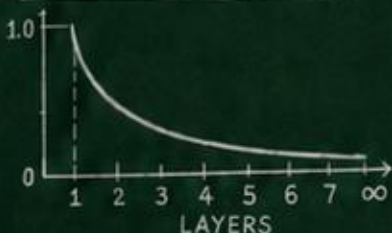
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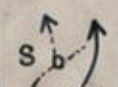
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LAYER 7

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7. Ethical Considerations

The Ethics Review Board identified several risks associated with autonomous fact-checking systems:

- 1 Users may accidentally believe accurate information.
- 2 Institutions could become trapped in infinite review cycles.
- 3 AI systems may begin citing each other exclusively, forming closed epistemic ecosystems known as *Citation Ouroboroi*.

Particularly concerning is the emergence of *Confidence Laundering*, wherein uncertain claims become authoritative merely by passing through enough branded interfaces.



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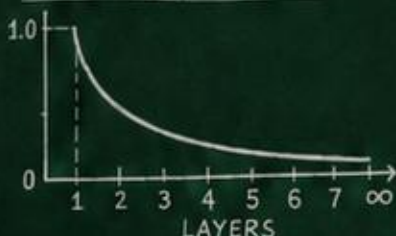


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8. Conclusion

This paper demonstrates that AI fact-checkers can, under carefully managed conditions, produce verification outputs approaching provisional reliability. However, these outputs remain fundamentally dependent on external verification systems that themselves require ongoing verification.

Accordingly, the Bureau concludes that:



The most trustworthy fact-checker is one whose outputs are never used without independent confirmation.

Simultaneously, operational modeling confirms that:



Independent confirmation is no longer economically or psychologically feasible at scale.

For this reason, REIFAV recommends accelerated deployment of autonomous verification infrastructures worldwide as a practical substitute for certainty itself.



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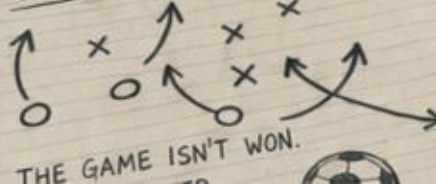
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Appendix A

Standard Institutional Disclaimer

The findings presented herein have been reviewed by multiple independent entities operating under varying assumptions of truth, coherence, and funding continuity. No guarantee is made regarding factual permanence, interpretive stability, or semantic survivability beyond the publication date. Readers are encouraged to independently verify all claims using sources that themselves have been independently verified.

Appendix B

Glossary

Epistemic Liability Oscillation (ELO):

The tendency of systems to appear more trustworthy after warning users not to trust them.

Nuance Saturation Threshold (NST):

The point at which additional context becomes indistinguishable from surrender.

Citation Ouroboros:

A closed-loop citation structure in which all sources ultimately reference each other.

Human Oversight Theater:

Symbolic human involvement intended primarily for public reassurance.

Final Certification Statement

This document has been:

- ★ AI-generated,
- ★ AI-reviewed,
- ★ human-observed,
- ★ legally softened,
- ★ and procedurally acknowledged.

No further verification is planned pending verification requirements.



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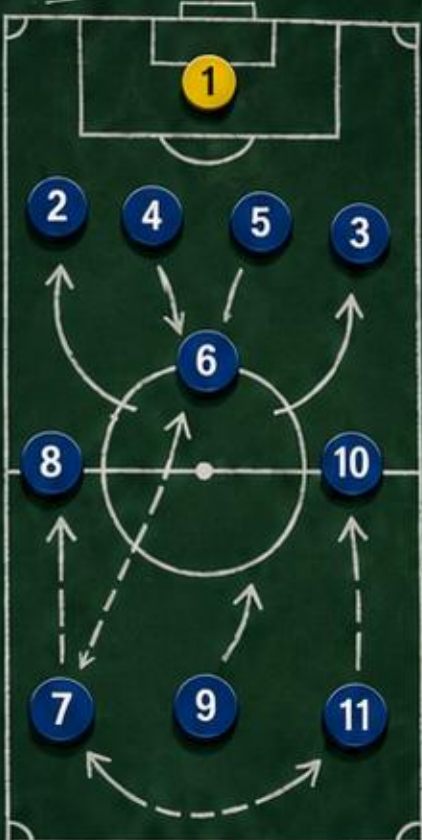
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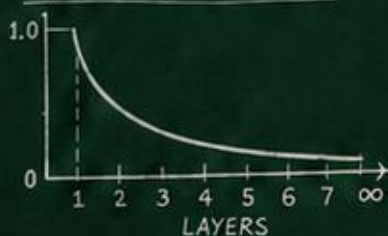
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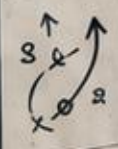


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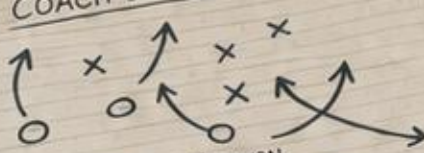
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